

NANYANG TECHNOLOGICAL UNIVERSITY

Lecture 6

EE3010: Electrical Devices and Machines

School of Electrical and Electronic Engineering

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Learning Objectives

By the end of this lecture, you should be able to:

- Describe the basic structure of a DC machine.
- Explain how voltage is induced in a rotating loop of wire and how the equation for induced voltage is derived in a DC machine.
- Explain the purpose of commutation.
- Explain how torque is produced when current is supplied to the loop, and how the equation for the induced torque is derived in a machine.

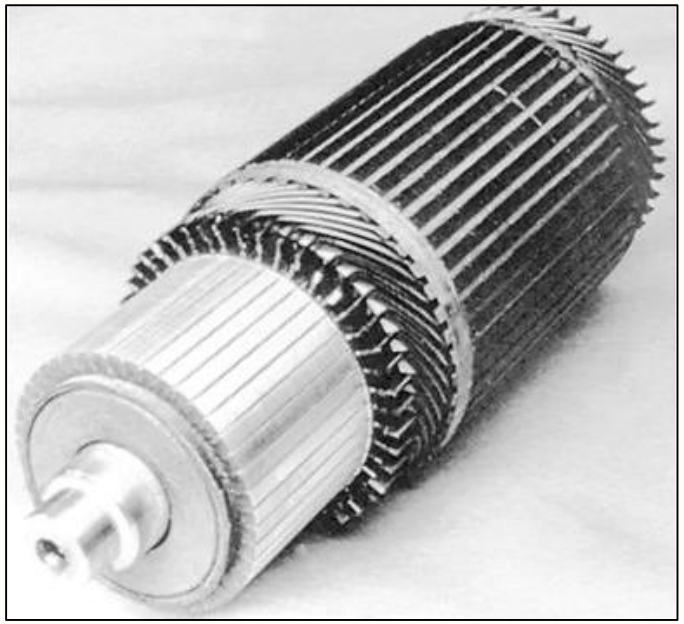
Introduction to DC Machinery Fundamentals

- DC machines are generators that convert mechanical energy to DC electric energy and motors that convert DC electric energy to mechanical energy.
- DC motors are used in a wide variety of industrial drives such as robots, machine tools, paper and steel mills.
- DC generators are quite rare in modern power systems. They are being replaced by solid-state devices that convert available AC power to DC power for DC drive systems and other DC applications.

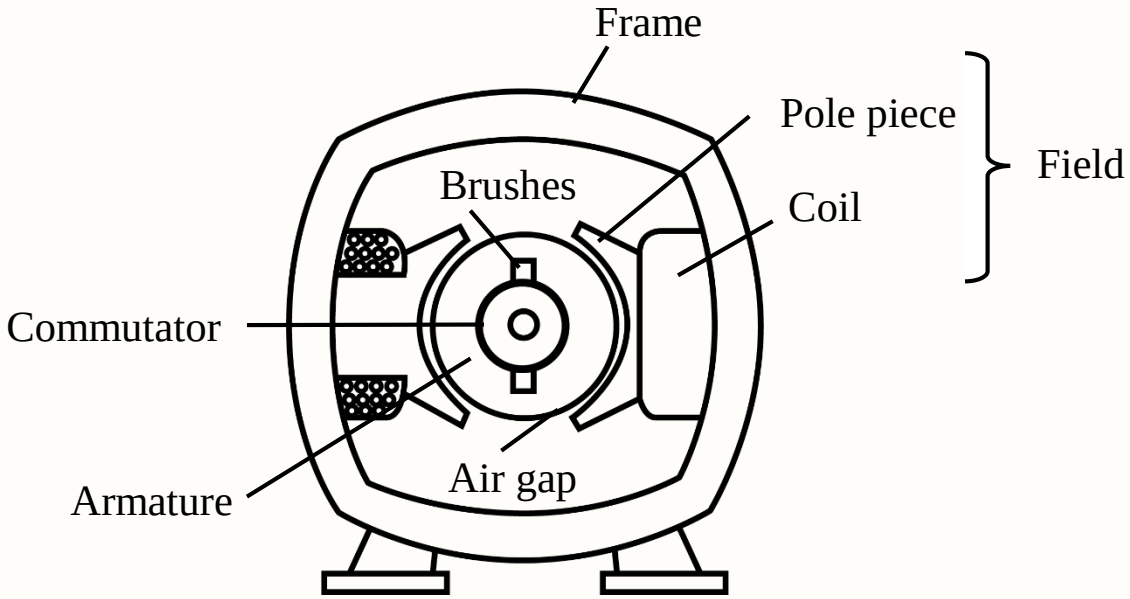
Introduction to DC Machinery Fundamentals

- However, DC generators have had a limited renaissance in last few years, as power sources for stand-alone cellular telephone towers.
- Most DC machines are like AC machines in that they have AC voltages and currents within them. A mechanism (commutators) exists, that converts the internal AC voltages to DC voltages at the terminals.
- Main parts of a DC machine are:
  - Stator – stationary part, consisting of the frame and pole pieces
  - Rotor – rotating part

Introduction to DC Machinery Fundamentals



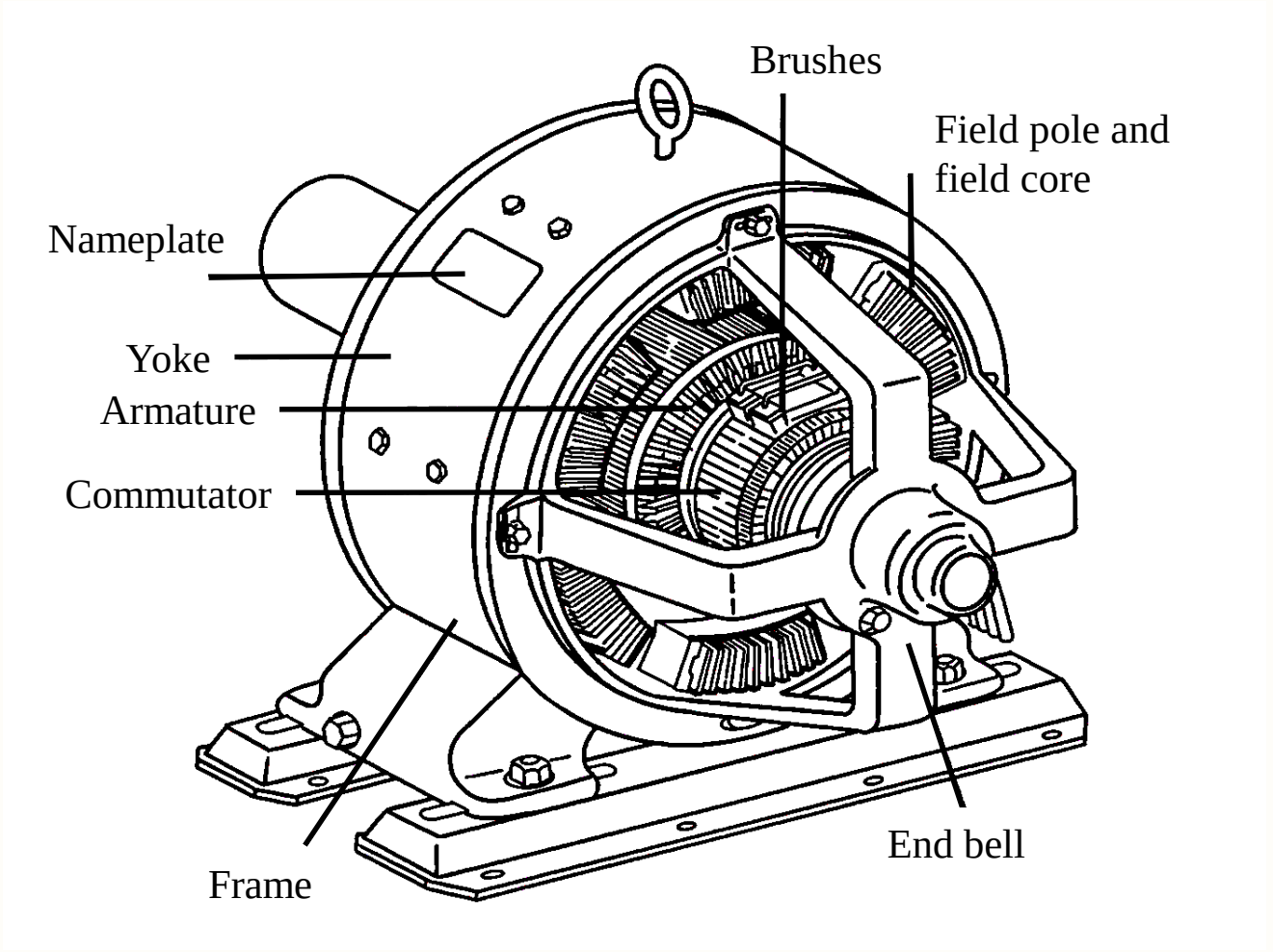
Armature of a DC generator. (*Used with courtesy of General Electric Company*) *WILDI, THEODORE, ELECTRICAL MACHINES, DRIVES AND POWER SYSTEMS, 6th, 2006*. Reprinted by permission of Pearson Education, Inc., New York, New York.



Cross section of a two-pole generator.

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Introduction to DC Machinery Fundamentals



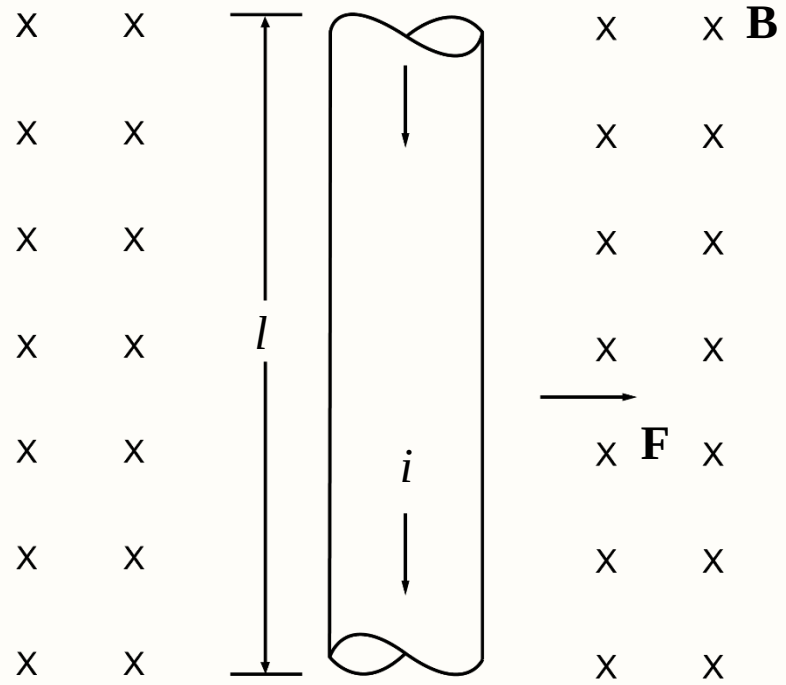
A simplified diagram of a DC machine.

Introduction to DC Machinery Fundamentals

- Two principal windings on a DC machine:
  - Armature windings - located on the rotor, where voltages are induced. Hence, a DC machine's rotor itself is sometimes called an armature.
  - Field windings - located on the stator, excited by DC currents to produce main magnetic flux in the machine.
- The fundamental principles involved in the operation of DC machines are very simple. Unfortunately, they are usually somewhat obscured by the complicated construction of real machines.

Fleming's Left Hand Rule (Motor)

- Whenever a current carrying conductor comes under a magnetic field, there is a force (Lorentz force) acting on the conductor. The force induced on the conductor is:



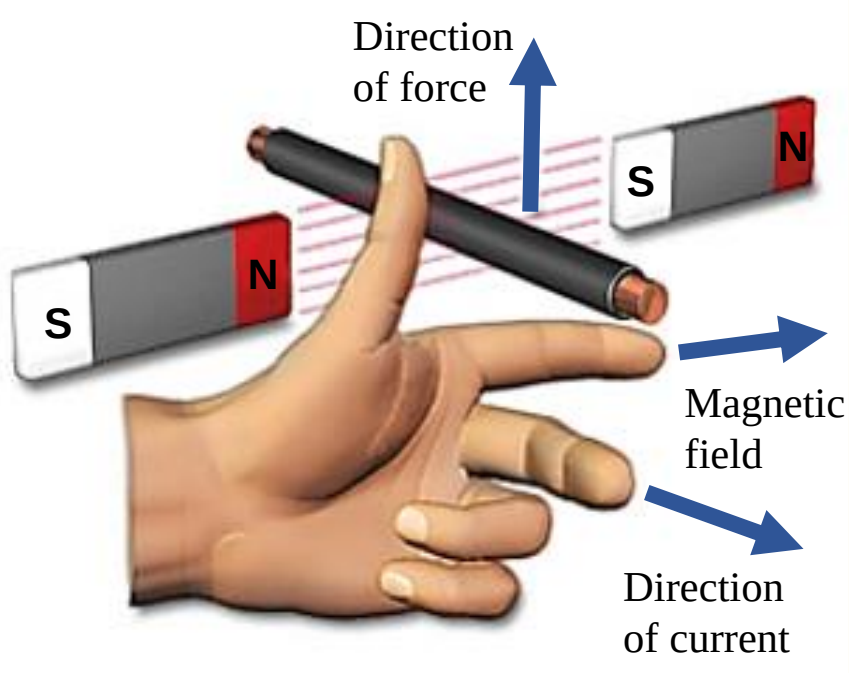
Force on a current-carrying wire in a magnetic field.

$$F = Bli$$

where

i = magnitude of the current
l = length of wire
B = magnetic flux density

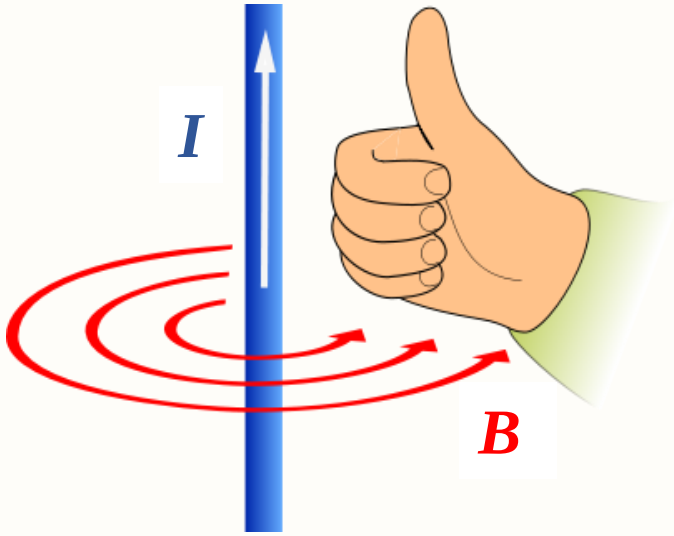
Fleming's Left Hand Rule (Motor)



Fleming's left hand rule.

- Hold out your left hand with forefinger, second finger and thumb at right angle to one another.
- If the forefinger represents the direction of the field and the second finger that of the current, then the thumb gives the direction of the force.

Maxwell's Right Hand Grip Rule

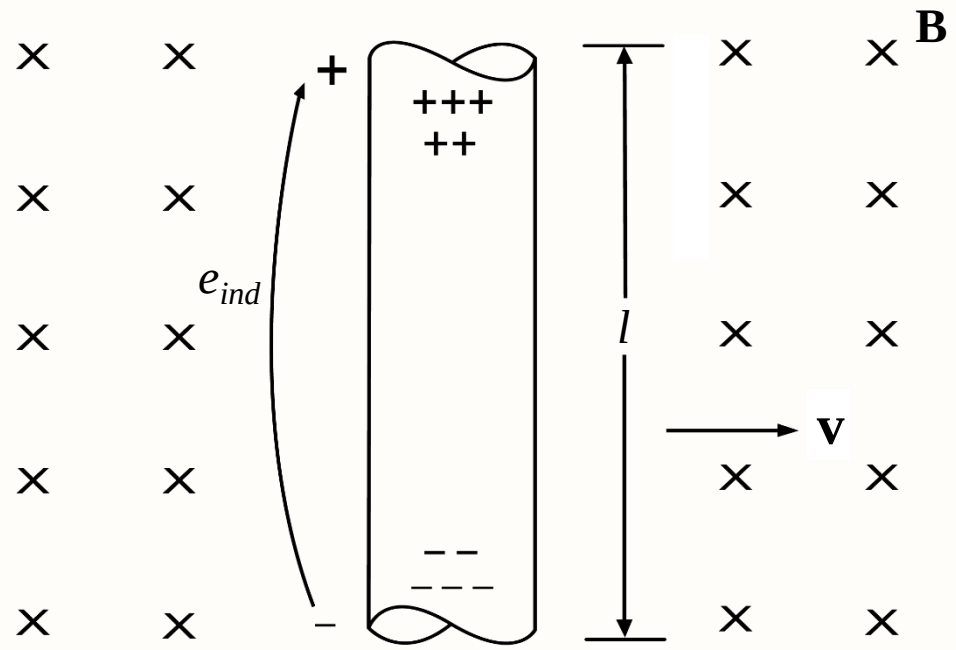


Maxwell's right hand grip rule.

- When electric current flows through a conductor, a magnetic field is induced around it.
- The direction of magnetic lines of force can be determined by Maxwell's right-hand grip rule.

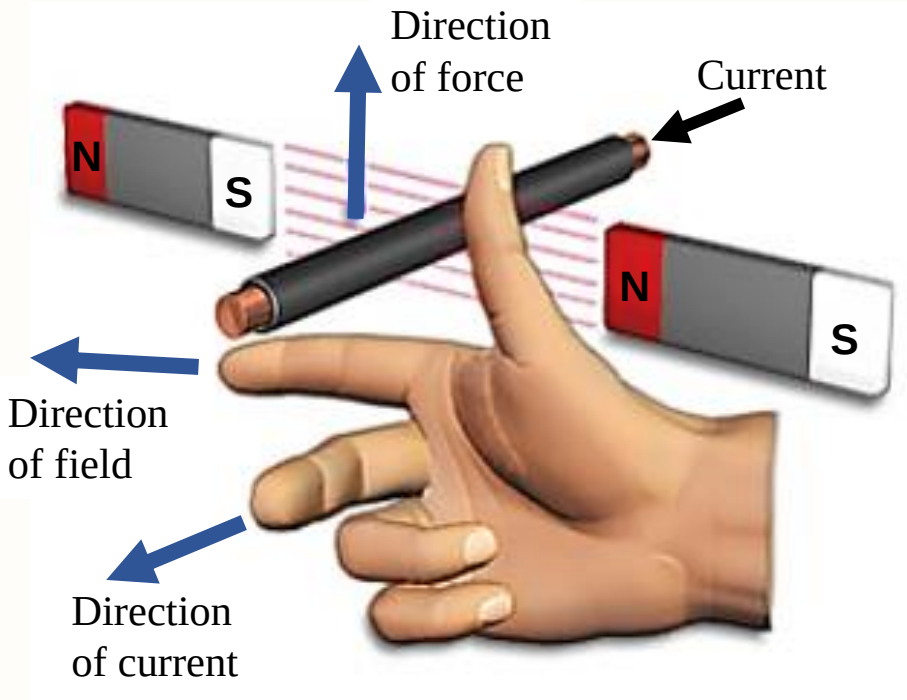
Fleming's Right Hand Rule (Generator)

- When a conductor of length l is moving at a velocity v to the right in a magnetic field of density B, according to Faraday's Law of Electromagnetic Induction, an emf e_{ind} will be induced.



$$e_{ind} = Blv$$

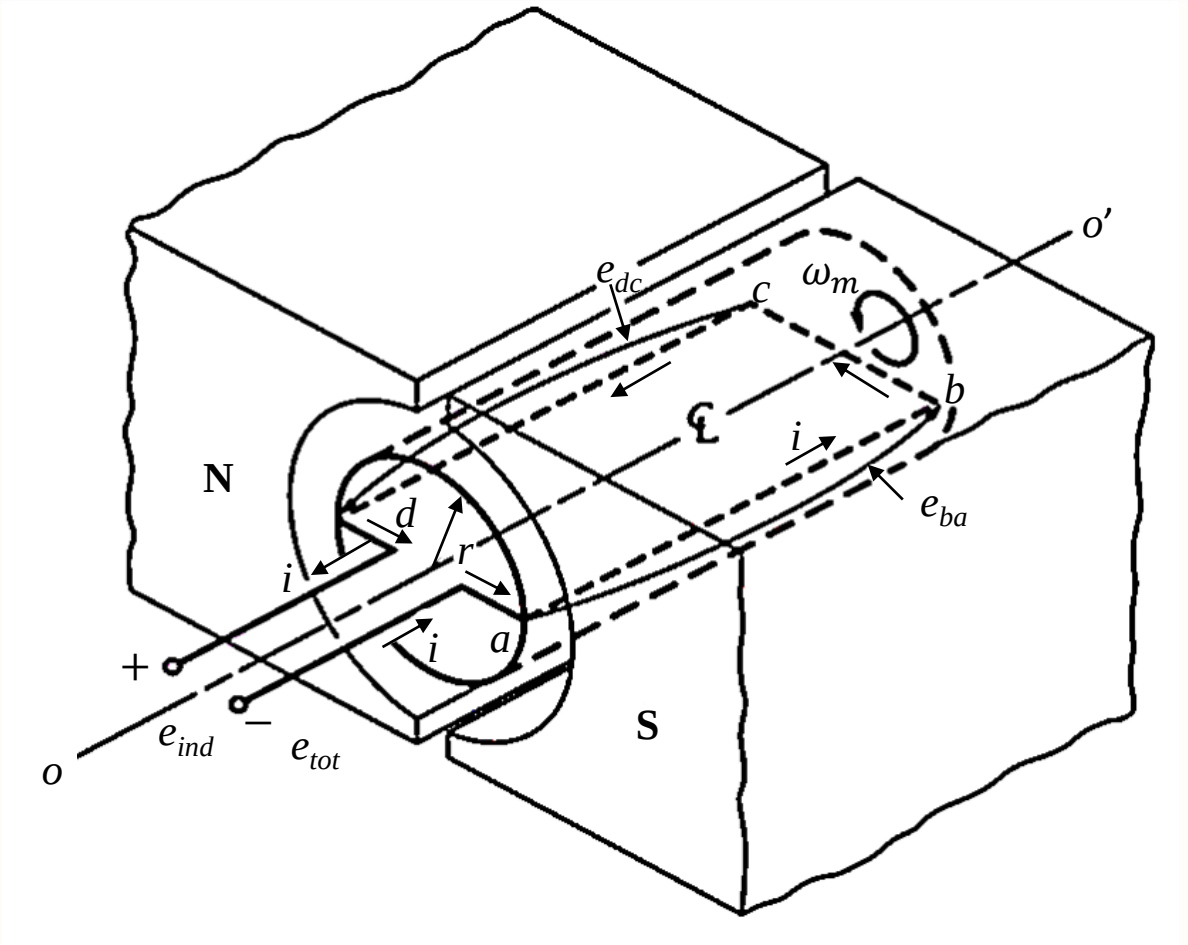
Fleming's Right Hand Rule (Generator)



Fleming's right hand rule.

- Hold out the right hand with the forefinger, second finger and thumb at right angle to each other.
- If forefinger represents the direction of the line of field, the thumb points in the direction of motion or applied force, then the second finger points in the direction of the induced current.

Principles of DC Machine Operations



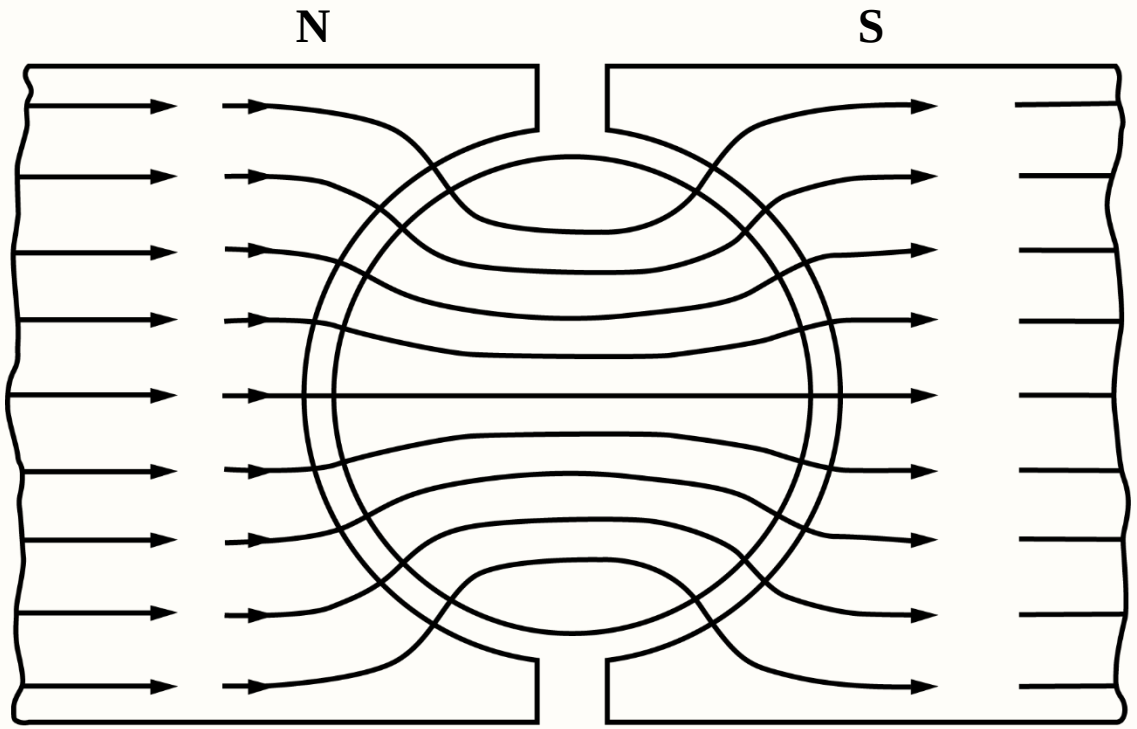
Simplest possible rotating DC machine: a simple rotating loop between curved pole faces.

Principles of DC Machine Operations

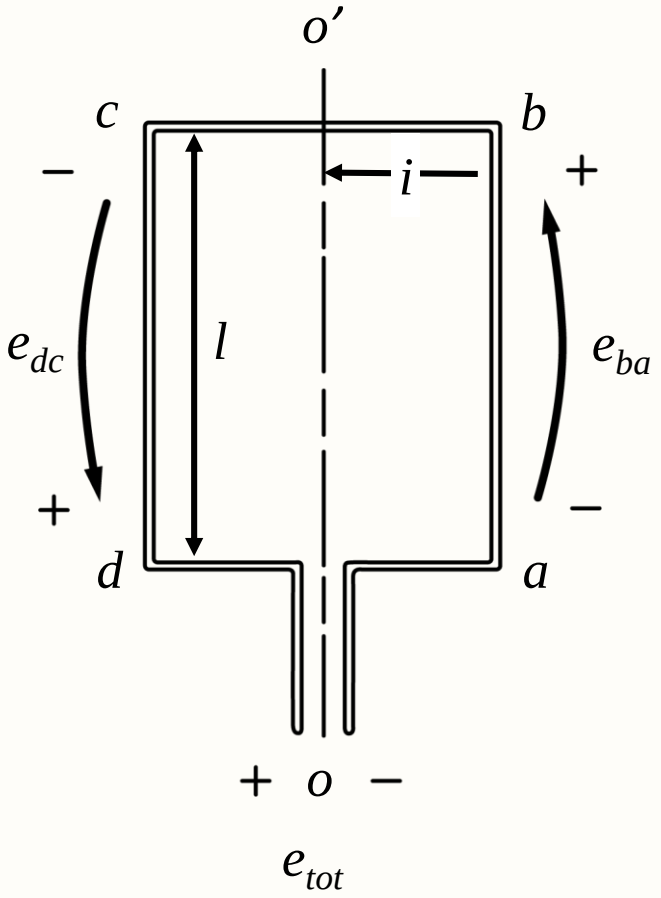
- The simplest possible rotating DC machine consists of a single loop of wire which lies in a slot carved in a ferromagnetic core and rotating about a fixed axis. The magnetic field is supplied by the magnetic north and south poles as shown.

- Assumptions:
  - Constant-width air gap between rotor and stator.
  - Air gap flux density is radial.
  - The flux density is constant everywhere under the poles faces and rapidly falls to zero beyond the edges of the poles.

The Voltage Induced in a Rotating Loop



View of field lines.



Top view.

The Voltage Induced in a Rotating Loop

- As the rotor is rotated, a voltage will be induced in the wire loop.
- To determine the total voltage induced on the loop, examine each segment of the loop separately and sum all the resulting voltages. The velocity of the wire is tangential to the path of rotation.

a) Segment *ab.*

$$e_{ba} = \begin{cases} Blv & \text{positive into page, under pole face} \\ 0, & \text{beyond the pole edges} \end{cases}$$

The Voltage Induced in a Rotating Loop

b) Segments *bc* and *da*.

They are the end connections used to connect the conductor segments *ab* and *cd* in series. The voltages induced are zero.

$$e_{cb} = 0, \; e_{ad} = 0$$

c) Segment *cd.*

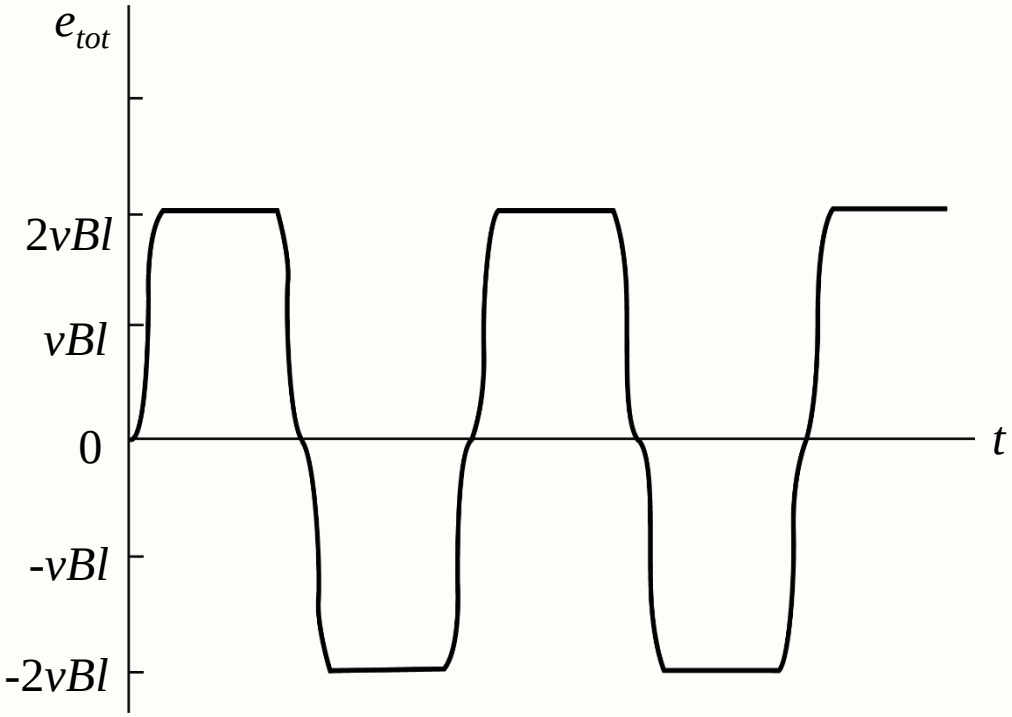
$$e_{dc} = \begin{cases} Blv & \text{positive out of page, under pole face} \\ 0, & \text{beyond the pole edges} \end{cases}$$

The total induced voltage in the loop is given by:

$$e_{ind} = e_{ba} + e_{ad} + e_{dc} + e_{cb} = \begin{cases} 2Blv, & \text{under the pole faces} \\ 0, & \text{beyond the pole edges} \end{cases}$$

The Voltage Induced in a Rotating Loop

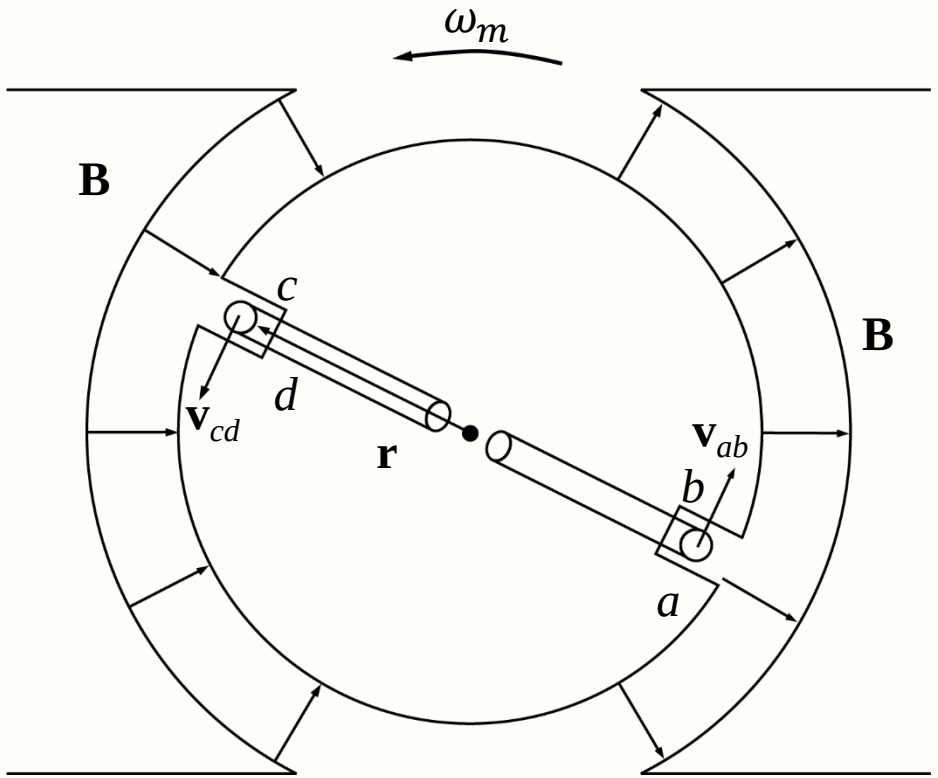
- When the loop rotates through 180 degrees, segment *ab* is under the north pole face instead of the south pole face. At that time, the direction of the voltage on the segment reverses, but its magnitude remains constant. The resulting total output voltage is as shown.



The total output voltage waveform of the loop.

Alternative Way to Express Total Induced Voltage

- To clearly relate the behaviour of the single loop to the behaviour of the larger, real DC machines, the tangential velocity v of the loop can be expressed as:



$$v = r\omega_m$$

where

r = radius from axis of rotation out to the edge of loop

ω_m = angular velocity of loop

Alternative Way to Express Total Induced Voltage

$$e_{ind} = \begin{cases} 2Blr\omega_m = \dfrac{2}{\pi} A_p B \omega_m = \dfrac{2}{\pi} \phi \omega_m, & \text{under the pole faces} \\ 0, & \text{beyond the pole edges} \end{cases}$$

Where $\begin{cases} \phi = A_p B \\ A_p = \pi r l, \text{ is the rotor surface area under each pole} \end{cases}$

Thus, the voltage generated in a DC machine is directly proportional to the field inside the machine and the speed of rotation of the machine. In general,

$$e_{ind} = \begin{cases} k\phi\omega_m & \text{under the pole faces} \\ 0 & \text{beyond the pole edges} \end{cases}$$

k is a constant representing the mechanical construction of the machine.

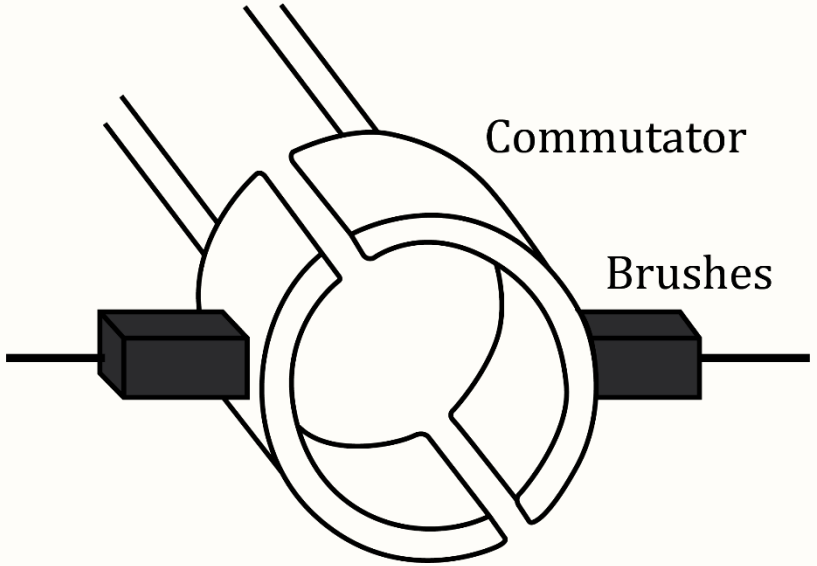
Getting DC Voltage out of Rotating Loop: Commutation

- Two semi-circular conducting segments are added to the end of the loop, and two fixed contacts are set up, such that at the instant when the voltage in the loop is zero, the contacts short-circuit the two segments. Hence, every time the voltage of the loop switches directions, the contacts also switch connections, and the output of the contacts is always built up in the same way (DC).

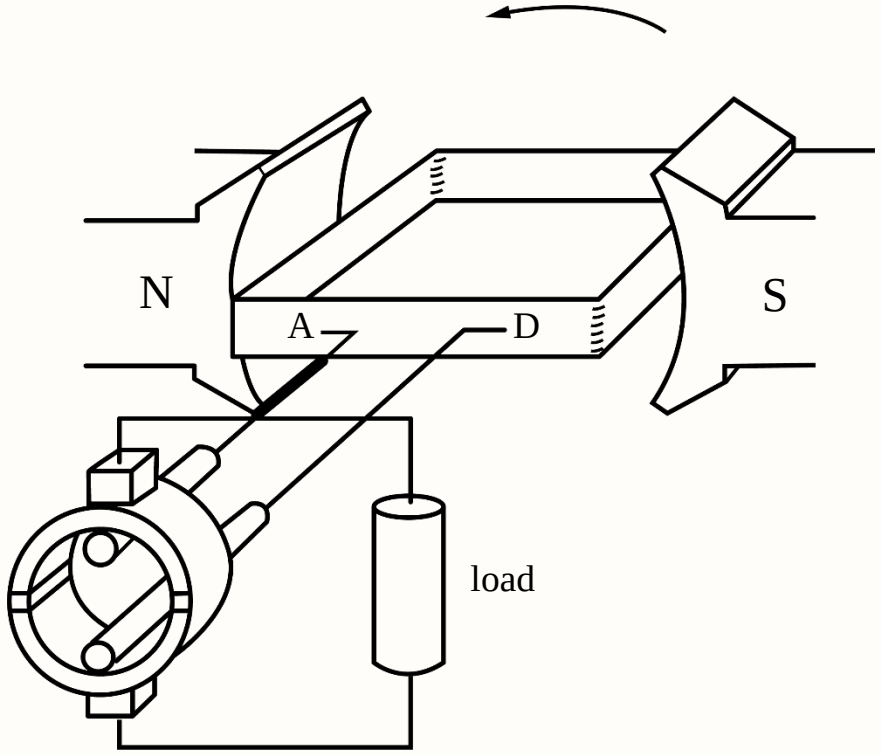
- This connection-switching process is known as commutation.

Getting DC Voltage out of Rotating Loop: Commutation

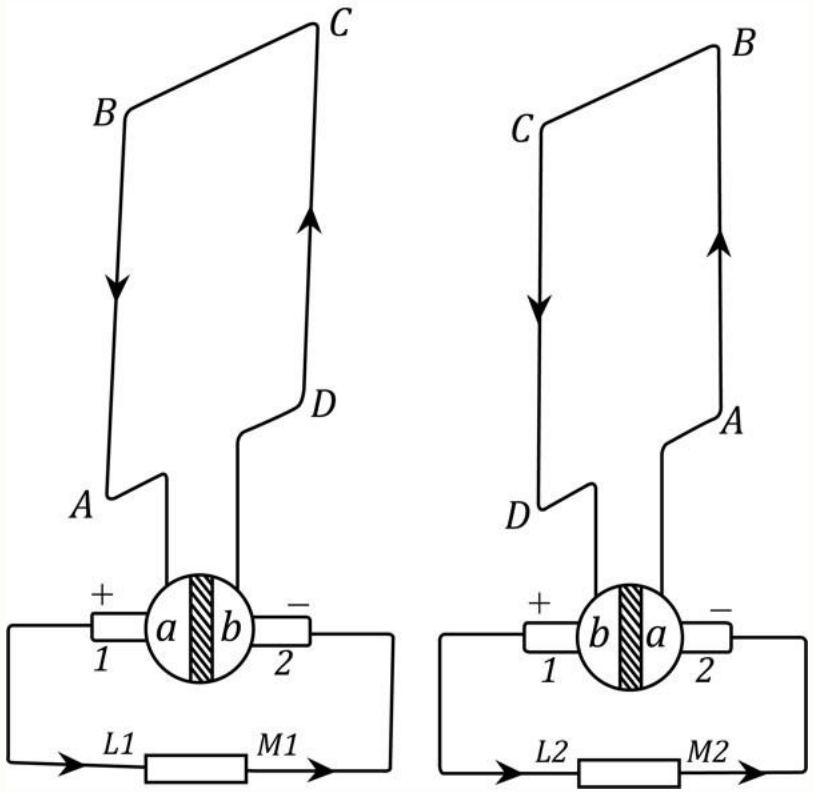
- The rotating semi-circular segments are called commutator segments, and the fixed contacts are called brushes. The brushes are pressed against the commutator by springs.



Getting DC Voltage out of Rotating Loop: Commutation



Elementary DC generator is simply an AC generator equipped with a mechanical rectifier called a commutator. *WILDI, THEODORE, ELECTRICAL MACHINES, DRIVES AND POWER SYSTEMS, 6th, 2006*. Reprinted by permission of Pearson Education, Inc., New York, New York.

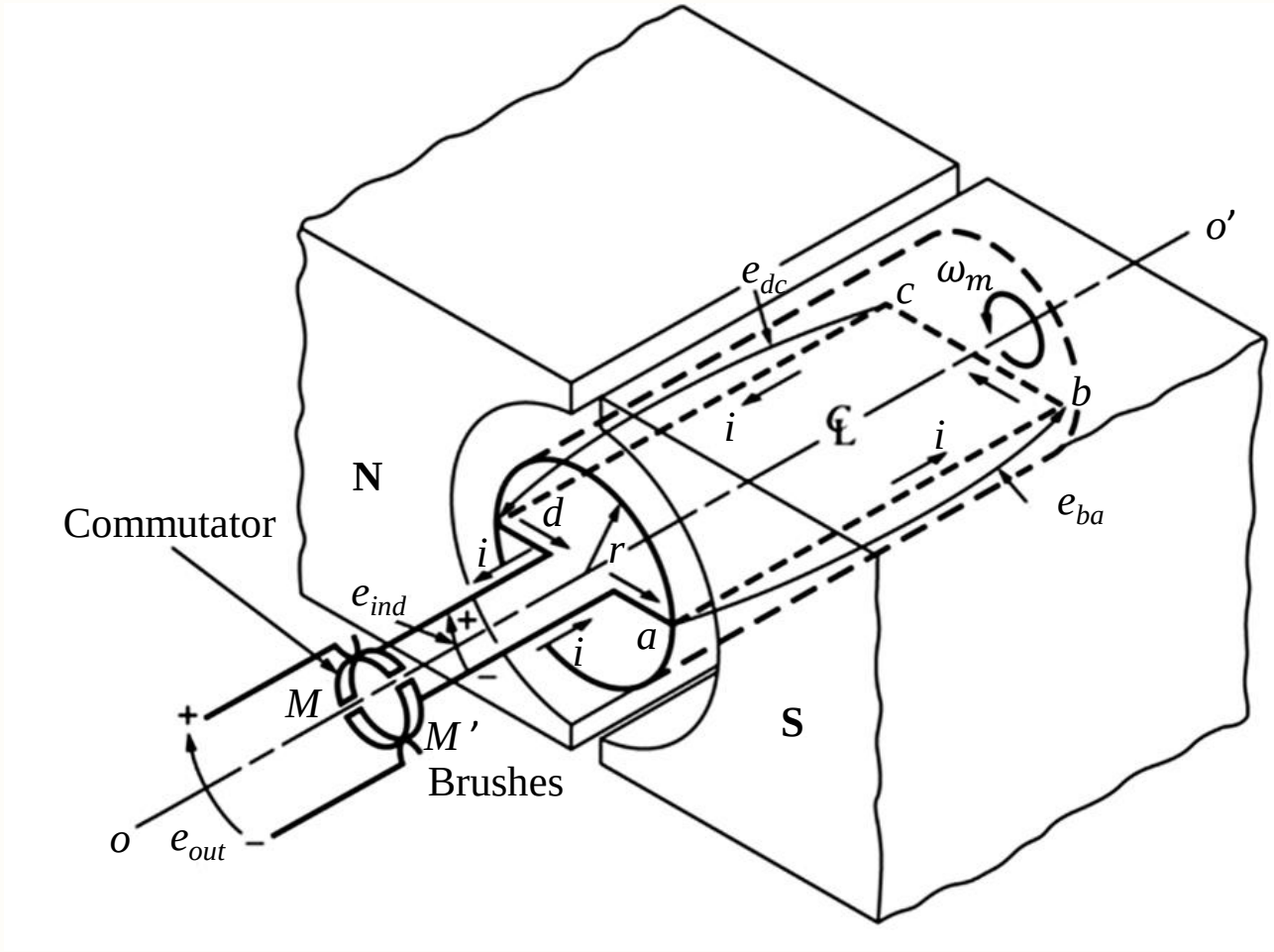


Principle of commutation.

Getting DC Voltage out of Rotating Loop: Commutation

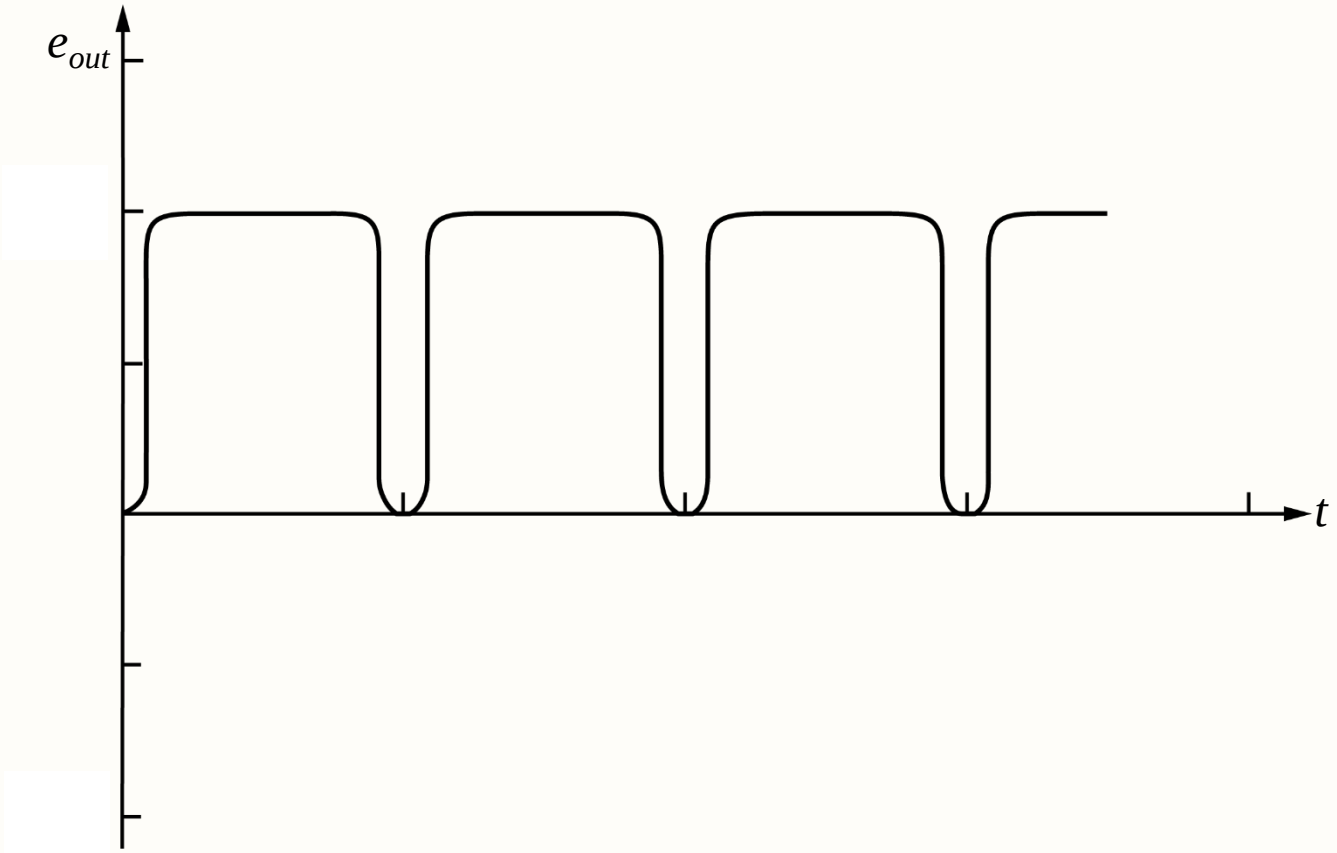
- A commutator in its simplest form is composed of a slip ring that is cut in half, with each segment insulated from one another as well as from the shaft. One segment is connected to each end of the loop.
- The commutator revolves with the loop and the voltage between the segments is picked up by two stationary brushes.
- The voltage between the brushes never changes polarity.
- The AC voltage in the loop is rectified by the commutator, which actually acts as a mechanical rectifier.

Getting DC Voltage out of Rotating Loop: Commutation



Producing a DC output with commutator and brushes.

Getting DC Voltage out of Rotating Loop: Commutation

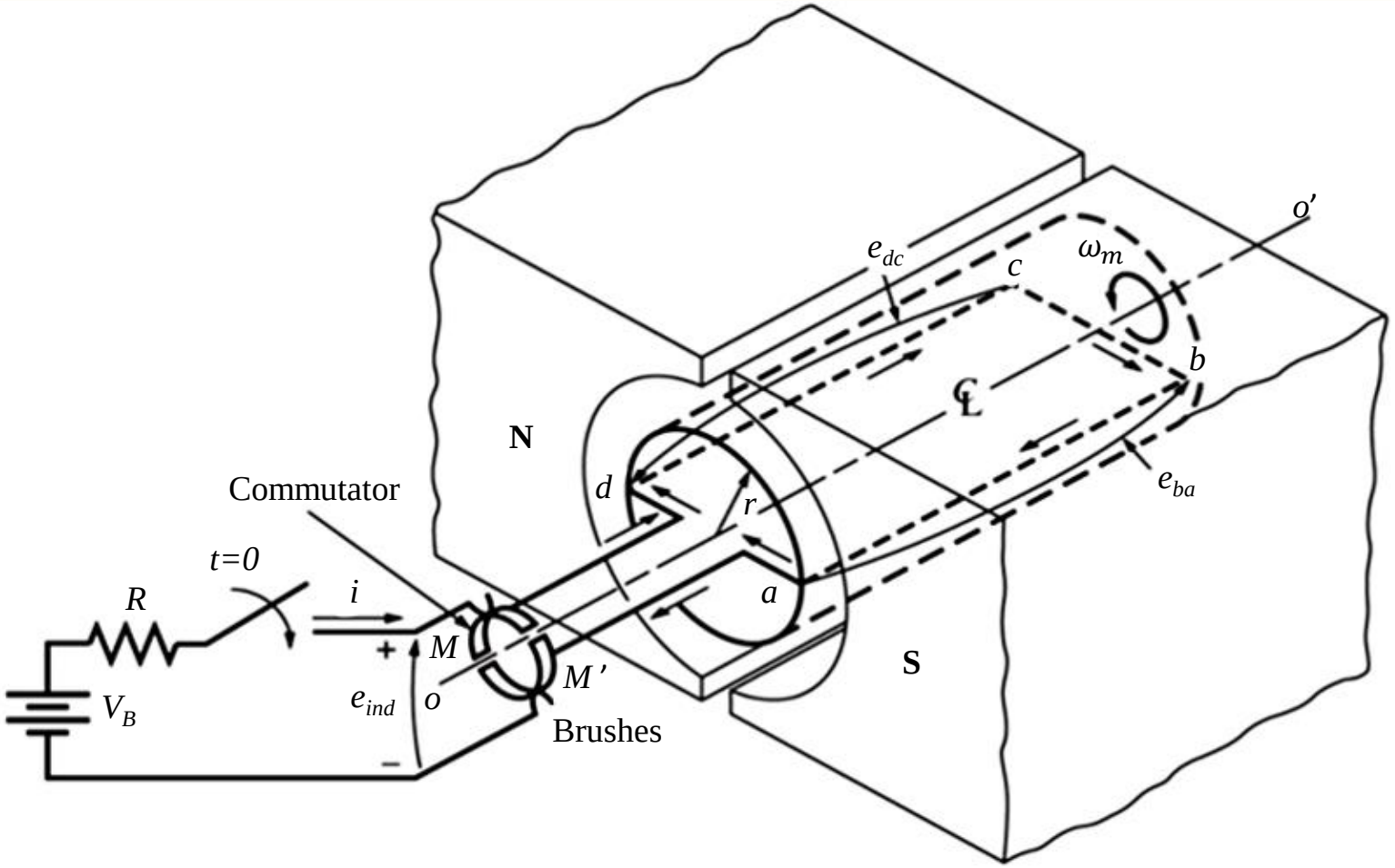


Resulting output voltage after commutation.

The Induced Torque in the Rotating Loop

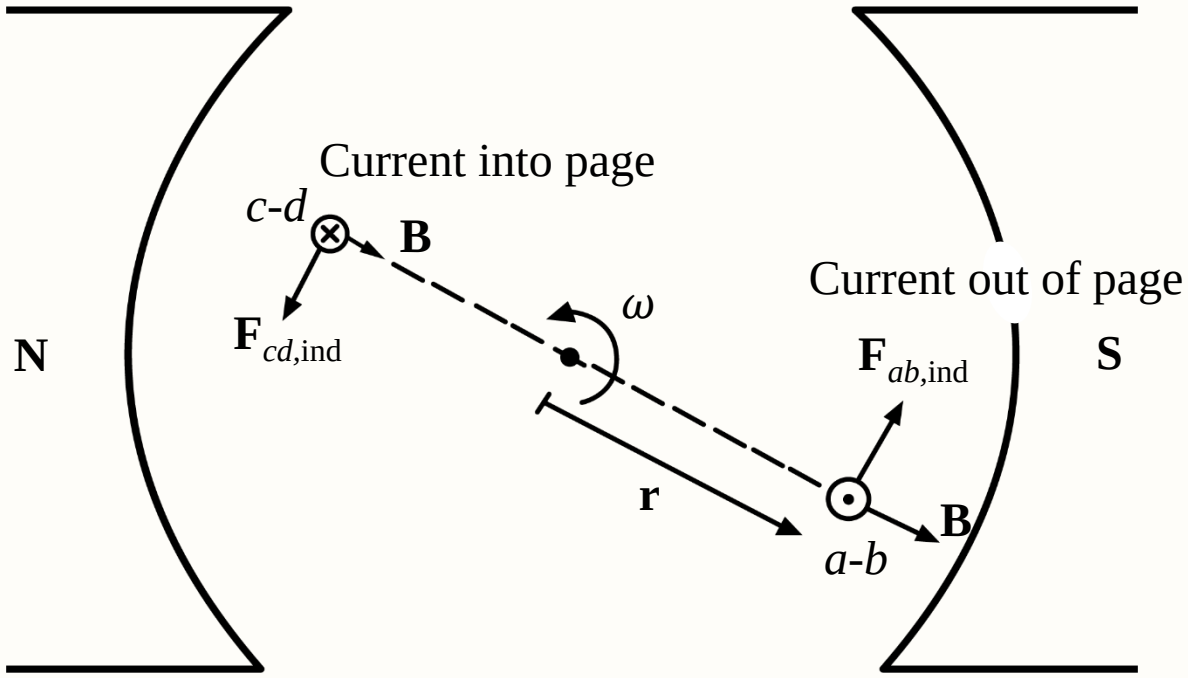
- Suppose a battery is connected to the machine as shown.
- The current that flows in the loop will result in a torque which rotates the loop.
- As the loop turns, the current in its windings is commutated to produce a continuous torque output.
- How much torque will be produced in the loop when the switch is closed and a current is flowing into it?
- The approach is to look at one segment of the loop at a time, and then sum the effects of all the individual segments.

The Induced Torque in Rotating Loop



A simple form of DC motor with commutator.

The Induced Torque in Rotating Loop



a) Segment *ab*.

$F_{ab} = Bli$, tangent to the direction of motion

$T_{ab} = F_{ab} r = Blir$, counter clockwise

The Induced Torque in Rotating Loop

b) Segments *bc* and *da*.

They are the end connections used to connect the conductor segments *ab* and *cd* in series. $T_{bc} = T_{da} = 0$

c) Segment *cd.*

$F_{cd} = Bli,$ tangent to the direction of motion

$T_{cd} = F_{cd} r = Blir,$ counter clockwise

The total induced torque on the loop is given by:

$$T_{ind} = T_{ab} + T_{bc} + T_{cd} + T_{da} = \begin{cases} 2Blir, & \text{under the pole faces} \\ 0, & \text{beyond the pole edges} \end{cases}$$

The Induced Torque in Rotating Loop

$$T_{ind} = \begin{cases} \frac{2}{\pi}\phi i, & \text{under the pole faces} \\ 0, & \text{beyond the pole edges} \end{cases}$$

$$\phi = A_p B, \quad A_p = \pi r l$$

Thus, the torque produced in the machine is directly proportional to the field (flux) in the machine and the rotor (armature) current. In general,

$$T_{ind} = \begin{cases} k\phi i, & \text{under the pole faces} \\ 0, & \text{beyond the pole edges} \end{cases}$$

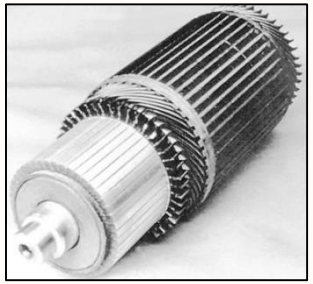
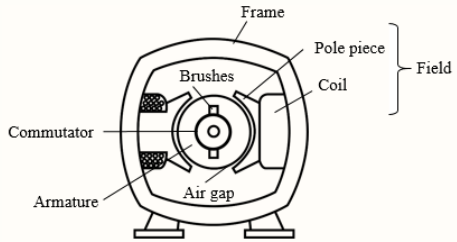
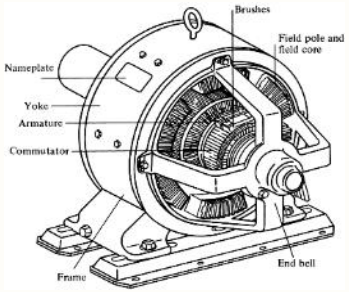
k is a constant representing the construction of the machine.

Summary

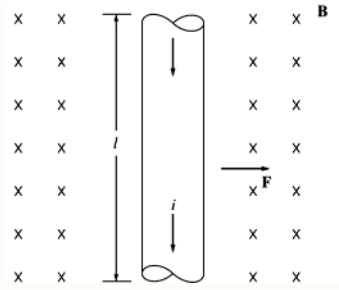
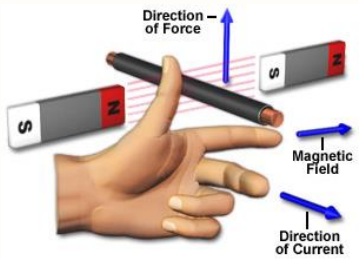
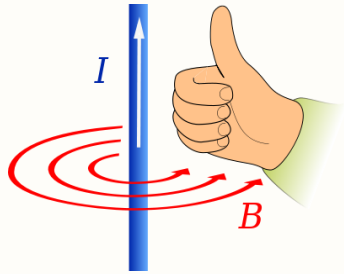
In this lecture, you have learnt:

- An internal voltage is generated when the rotor is rotated in the magnetic field created by the stator.
- The internal generated emf equation that shows that it is directly proportional to the field in the machine and the speed of rotation of the machine.
- A torque is produced when the rotor is supplied with a DC current and under the influence of the magnetic field set up by the stator circuit.
- The torque equation that shows that it is directly proportional to the field in the machine and the rotor current.
- The concept of commutation that converts AC voltages in the rotor of a DC machine to DC voltage at its terminals.

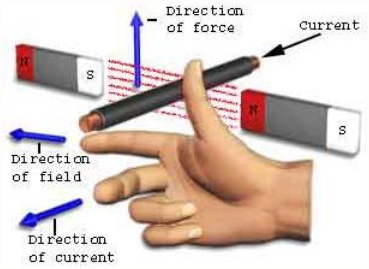
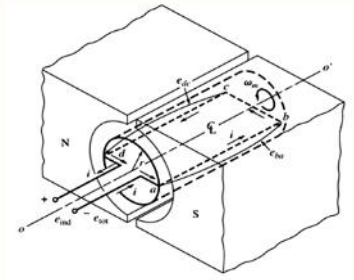
References

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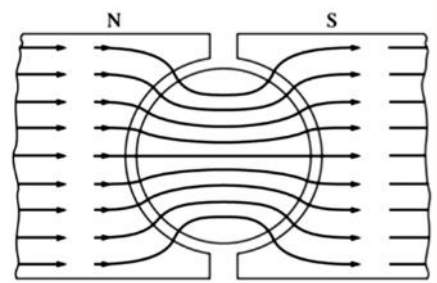
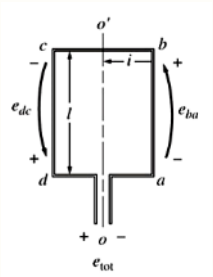
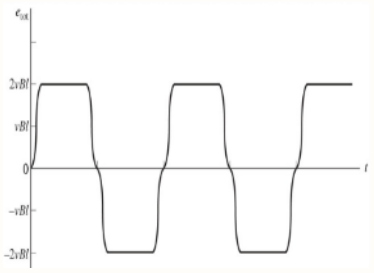
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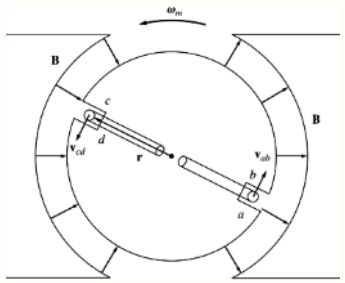
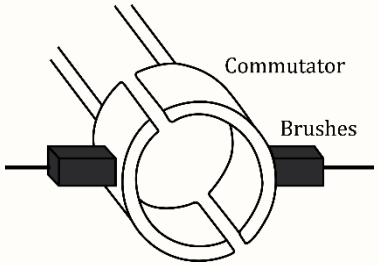
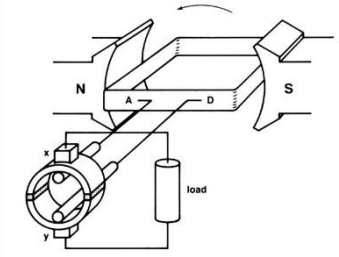
References

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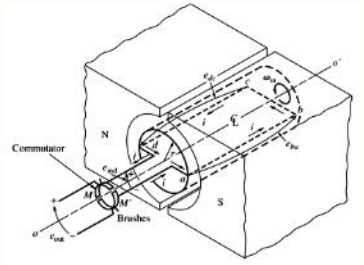
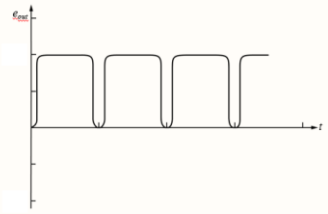
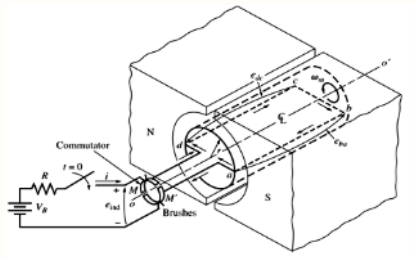
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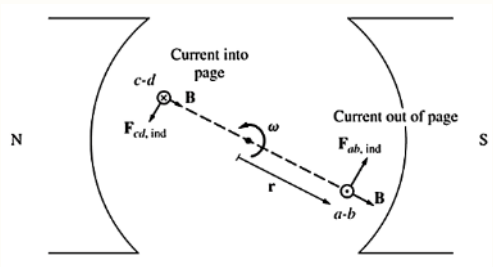
References

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